Exp No: 10 Experiment to understand array function in Data science.

AIM:

To conduct an experiment to understand array function in Data science.

PROGRAM:

import numpy as np

array = np.random.randint(1, 100, 9)

array

```
array([[30, 76, 68],
       [70, 39, 58]])
```

np.sqrt(array)

```
array([7.        , 9.        , 2.44948974, 2.64575131, 3.60555128,
       5.        , 2.82842712, 8.83176087, 3.        ])
```

array.ndim

```
1
```

new_array = array.reshape(3, 3)

new_array

```
array([[49, 81,  6],
       [ 7, 13, 25],
       [ 8, 78,  9]])
```

new_array.ndim

```
2
```

new_array.ravel()

```
array([49, 81,  6,  7, 13, 25,  8, 78,  9])
```

newm = new_array.reshape(3, 3)

newm

```
array([[92, 25, 65],
       [20, 52, 32],
       [ 6, 10, 39]])
```

newm[2, 1:3]

```
  array([10, 39])
```

newm[1:2, 1:3]

```
array([[13, 25]])
```

new_array[0:3, 0:0]

```
array([], shape=(3, 0), dtype=int64)
```

new_array[0:2, 0:1]

```
array([[49],
       [ 7]])
```

new_array[0:3, 0:1]

```
array([[49],
       [ 7],
       [ 8]])
```

new_array[1:3]

```
array([[ 7, 13, 25],
       [ 8, 78,  9]])
```

RESULT:

Thus an experiment to understand array function in Data science was conducted.